MSE-2 CASE STUDY REPORT

E-Waste & Circular Economy Case Studies

Prepared for MSE-2 Examination

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1. Japan Recycles More Than Most Countries: Lessons for India

Executive Summary

Japan has developed a highly organized waste-management system based on waste segregation, reduction, recycling, municipal responsibility and public participation. Japan's FY2024 municipal waste data shows a waste-reduction rate of 99.2%, direct landfill of 0.8%, and a recycling rate of 19.3%.

Background

- Japan has limited land availability for waste disposal.
- Waste management emphasizes reduction and controlled treatment.
- Municipalities organize collection and treatment.
- Citizens separate waste into specified categories.
- Japan has extensive waste-to-energy infrastructure.

Strategy / Initiative

1. Source segregation.

2. Separate collection of recyclable materials.

3. Reduction of waste generation.

4. Recycling and resource recovery.

5. Incineration of residual waste.

6. Energy recovery from waste.

7. Public participation and awareness.

Impact & Results

- Total municipal waste: approximately 38.5 million tonnes.
- Waste-reduction rate: 99.2%.
- Direct landfill rate: 0.8%.
- Recycling rate: 19.3%.
- Incineration plants: 991.
- Plants with electricity generation: 415.

SWOT Analysis

Strengths: Strong segregation, high treatment capacity, public participation.

Weaknesses: High infrastructure cost, dependence on incineration, strict compliance requirements.

Opportunities: Advanced recycling, waste-to-energy, application of Japanese practices in India.

Threats: Increasing consumption, ageing infrastructure, high operating costs.

Comparison with India

Japan has stronger source segregation, highly organized municipal systems, very low direct landfill and extensive waste-treatment infrastructure. India is still expanding collection, segregation and treatment capacity.

Recommendations

- Improve household segregation.
- Introduce stronger local collection systems.
- Encourage recycling before disposal.
- Use waste-to-energy for suitable residual waste.
- Improve citizen awareness.

Conclusion

Japan demonstrates that effective waste management requires technology, government systems and citizen participation.

One-Page Exam Revision Sheet

Keywords: Segregation → Collection → Recycling → Incineration → Energy Recovery → Public Participation.

Key figure: 99.2% waste-reduction rate.

Main lesson: Organized segregation and treatment reduce landfill dependence.

2. Switzerland's E-Waste Success Story

Executive Summary

Switzerland operates structured take-back systems such as SENS and SWICO, supported by collection points, retailers and financing mechanisms. The SENS system processed 100,762 tonnes in 2024, equivalent to 11.14 kg per person.

Background

- Switzerland has a long-established e-waste collection system.
- Consumers can return old electrical/electronic equipment.
- Retailers and collection points participate in take-back.
- Recycling costs are supported through an Advance Recycling Fee (ARF).

Strategy / Initiative

- Free take-back of old equipment.
- Dense collection network.
- Producer/importer participation.
- Professional dismantling.
- Recovery of valuable materials.
- Separate treatment of hazardous components.
- Advance Recycling Fee.

Impact & Results

- SENS processed volume, 2024: 100,762 tonnes.
- Per-capita amount: 11.14 kg/person.
- Collection through retailers and collection points.
- SENS and SWICO form major national take-back infrastructure.

SWOT Analysis

Strengths: Convenient collection, strong financing, professional recycling.

Weaknesses: High infrastructure cost and dependence on formal systems.

Opportunities: Higher material recovery and circular economy.

Threats: Increasing electronic consumption and rapidly changing technology.

Comparison with India

Switzerland has mature take-back systems, strong consumer convenience and established recycling financing. India's formal system is developing, with EPR certificates and a significant informal sector.

Recommendations

- Increase convenient collection points.
- Make producer take-back easier for consumers.
- Integrate informal workers into formal systems.
- Improve tracking of e-waste.

Conclusion

Switzerland shows that convenient collection, producer responsibility, reliable financing and professional recycling can create a strong e-waste ecosystem.

One-Page Exam Revision Sheet

Remember: SENS + SWICO + ARF + Take-back + Professional Recycling.

3. India's E-Waste Management Rules 2022

Executive Summary

India's E-Waste (Management) Rules, 2022 came into force on 1 April 2023 and introduced a stronger digital Extended Producer Responsibility (EPR) framework.

Background

India's increasing electronics consumption creates growing e-waste. The 2022 Rules aim to ensure environmentally sound management.

Strategy / Initiative

- Extended Producer Responsibility.
- Online EPR portal.
- Registered producers and recyclers.
- EPR recycling obligations.
- EPR certificates.
- Refurbishing certificates.
- Quarterly and annual returns.
- Environmental compensation for violations.
- Auditing of stakeholders.

Impact & Results

- Rules effective: 1 April 2023.
- EEE categories: 7.
- EEE types covered: 106.
- Registered e-waste producers/PIBOs reported in 2024: 7,050.

SWOT Analysis

Strengths: Digital tracking, producer responsibility, registered recyclers, EPR certificates.

Weaknesses: Enforcement challenges, informal-sector integration difficulties, variable reporting quality.

Opportunities: Formalization, traceability and increased material recovery.

Threats: Rapid electronics growth, illegal/informal processing and non-compliance.

Comparison with India

The 2022 Rules represent India's move from earlier approaches toward digital EPR and measurable recycling obligations.

Recommendations

- Improve enforcement.
- Integrate informal recyclers.
- Increase consumer awareness.
- Improve collection infrastructure.
- Strengthen auditing.

Conclusion

The 2022 Rules create a structured legal framework, but effectiveness depends on implementation, enforcement, collection and participation.

One-Page Exam Revision Sheet

EPR → Online Portal → Registered Recyclers → EPR Certificates → Recycling Targets → Returns → Environmental Compensation.

4. India's Extended Producer Responsibility (EPR): Success or Paper Compliance?

Executive Summary

EPR makes producers responsible for managing the environmental impact of products after their useful life. India's e-waste EPR system uses a digital portal where producers fulfil obligations through registered recyclers and EPR certificates.

Background

Producer → Product → Consumer → End-of-life → Collection → Recycler → Material Recovery.

Strategy / Initiative

- Producer registration.
- Recycling targets.
- Registered recyclers.
- EPR certificates.
- Digital reporting.
- Audits.
- Environmental compensation.

Impact & Results

- 7,050 registered e-waste producers were reported in 2024.
- The framework provides digital tracking and measurable obligations.
- Actual collection and recycling remain important implementation indicators.

SWOT Analysis

Strengths: Digital tracking, clear producer responsibility, registered recyclers, EPR certificates.

Weaknesses: Risk of paperwork-focused compliance, collection challenges and informal-sector integration.

Opportunities: Formal recycling growth, better traceability and circular economy.

Threats: Poor reporting, informal leakage and rapid e-waste growth.

Comparison with India

The key issue is the gap between policy design and implementation: the policy provides digital EPR, while actual collection, recycling and material recovery determine effectiveness.

Recommendations

- Independent audits.
- Digital tracking of collection.
- Stronger recycler verification.
- Consumer incentives.
- Formal integration of waste pickers.
- Public performance dashboards.

Conclusion

EPR should be evaluated not only by registration numbers but also by actual collection, recycling and material recovery.

One-Page Exam Revision Sheet

EPR = Producer Responsibility + Target + Recycler + Certificate + Monitoring.

5. Bengaluru — India's E-Waste Capital?

Executive Summary

Bengaluru's technology and electronics ecosystem generates substantial e-waste. A 2026 local report estimated approximately 12,000 tonnes of e-waste annually, with around 15% scientifically processed according to the cited report.

Background

- Bengaluru is a major IT and technology hub.
- Large numbers of computers and electronics become obsolete.
- Informal scrap channels play a major role.
- Karnataka has authorized recycling units.

Strategy / Initiative

- Authorized e-waste recyclers.
- Collection centres.
- Corporate take-back.
- Informal-sector collection.
- EPR-based formalization.

Impact & Results

- Estimated annual e-waste: ~12,000 tonnes.
- Scientifically processed: ~15% according to the cited report.
- Karnataka authorized recycling units: 108+ reported.

SWOT Analysis

Strengths: Technology ecosystem, recycling businesses and skilled workforce.

Weaknesses: Large informal sector and collection gaps.

Opportunities: IT-company take-back, digital tracking and refurbishment.

Threats: Increasing electronics consumption and unsafe informal processing.

Comparison with India

Bengaluru illustrates a national issue: technology-rich cities can generate significant e-waste while formal collection does not capture all waste.

Recommendations

- Establish collection points near IT clusters.
- Require corporate take-back programmes.
- Improve consumer awareness.
- Integrate informal collectors.
- Increase authorized processing capacity.

Conclusion

Bengaluru's case shows the gap between high e-waste generation and formal scientific processing.

One-Page Exam Revision Sheet

Bengaluru = IT Hub → High Electronics Use → High E-Waste → Informal Collection → Need Formal Recycling.

6. Mumbai's E-Waste Ecosystem

Executive Summary

Mumbai's e-waste ecosystem includes households, businesses, scrap dealers, informal collectors, authorized recyclers and government institutions.

Background

Mumbai has high population density, a large commercial sector, high electronics consumption, extensive informal recycling activity and authorized recycling facilities.

Strategy / Initiative

Collection → Segregation → Scrap Dealer/Collection Centre → Authorized Recycler → Material Recovery.

Impact & Results

Example: Mumbai Port Authority reported e-waste sent to authorized recyclers:

- 2022–23: 25.05 MT.
- 2023–24: 17.74 MT.
- 2024–25: 6.06 MT.
- 2025–26 up to 25 December 2025: 0.810 MT.

SWOT Analysis

Strengths: Large recycling ecosystem, established scrap networks and strong market for recovered materials.

Weaknesses: Informal operations and uneven traceability.

Opportunities: Formalization, digital collection and corporate take-back.

Threats: Unsafe processing and improper disposal.

Comparison with India

Mumbai represents a major metropolitan model where formal and informal recycling coexist.

Recommendations

- Formalize waste-picker networks.
- Increase authorized collection points.
- Digitalize collection records.
- Encourage corporate collection drives.
- Strengthen enforcement against unsafe processing.

Conclusion

Mumbai demonstrates the economic importance of recycling while highlighting the need to connect informal collection with environmentally sound formal recycling.

One-Page Exam Revision Sheet

Mumbai = Consumers + Businesses + Informal Collectors + Scrap Dealers + Authorized Recyclers.

7. Kerala's Clean Kerala Company

Executive Summary

Clean Kerala Company Limited (CKCL) is a Government of Kerala undertaking that provides forward linkages for local governments in solid-waste processing and disposal. It works with local governments and Haritha Karma Sena.

Background

CKCL focuses on segregated plastic, reject waste, e-waste, hazardous e-waste and glass waste.

Strategy / Initiative

1. Local collection.

2. Segregation.

3. Material Collection Facilities.

4. Linkage with recycling companies.

5. Safe disposal of hazardous materials.

6. Participation of Haritha Karma Sena.

Impact & Results

- Segregated plastic collected: 38,406 MT.
- E-waste collected: 2,678 MT.
- Hazardous e-waste: 205 MT.
- Glass waste: 6,429 MT.
- Rejects: 1,23,986 MT.

SWOT Analysis

Strengths: Government support, local-government network and organized collection.

Weaknesses: Processing capacity and market dependence.

Opportunities: Circular economy, recycling industries and green jobs.

Threats: Increasing waste generation and contamination.

Comparison with India

Kerala demonstrates how local governments, organized collection workers and centralized processing linkages can improve waste management.

Recommendations

- Expand collection coverage.
- Improve material recovery.
- Strengthen e-waste collection.
- Increase citizen participation.
- Develop local recycling markets.

Conclusion

CKCL demonstrates the value of creating a structured link between local collection and authorized processing.

One-Page Exam Revision Sheet

CKCL → Local Governments → Haritha Karma Sena → Collection → Processing → Recycling.

8. Pune's SWaCH Model

Executive Summary

SWaCH is a cooperative of self-employed waste collectors and urban poor that provides front-end waste-management services in Pune. Its reported 2025 figures include 1 million households covered daily, 4,000+ waste-pickers integrated and 82,891 tonnes of recycled waste annually.

Background

SWaCH integrates waste-pickers into organized municipal waste management through door-to-door collection and segregation.

Strategy / Initiative

- Door-to-door collection.
- Waste segregation.
- Waste-picker cooperative.
- Recycling.
- Composting.
- Citizen participation.
- Partnership with Pune Municipal Corporation.

Impact & Results

- Households covered daily: 10 lakh.
- Waste-pickers integrated: 4,000+.
- Waste recycled annually: 82,891 tonnes.
- Coverage: 70%+ of Pune.
- Reported savings to PMC: ₹113 crore.

SWOT Analysis

Strengths: Social inclusion, employment, decentralized collection and high recycling.

Weaknesses: Dependence on worker participation and municipal coordination.

Opportunities: Expansion to other cities.

Threats: Policy changes, unstable recyclable prices and inadequate segregation.

Comparison with India

SWaCH provides a model for Indian cities where informal waste-pickers can become part of formal waste management.

Recommendations

- Replicate cooperative models.
- Provide social security to waste-pickers.
- Improve source segregation.
- Establish stable municipal contracts.
- Use digital collection tracking.

Conclusion

SWaCH demonstrates that waste management can simultaneously create environmental and social benefits.

One-Page Exam Revision Sheet

SWaCH = Waste Pickers + Cooperative + Door-to-Door Collection + Recycling + Social Inclusion.

9. Dharavi Recycling Ecosystem

Executive Summary

Dharavi has developed a large informal recycling and manufacturing ecosystem. The network includes collection, sorting, trading, processing and manufacturing.

Background

Dharavi contains numerous small-scale recycling businesses dealing with plastic, paper, cardboard, metals, leather and other recyclable materials.

Strategy / Initiative

Waste Picker → Scrap Dealer → Trader → Sorting → Processing → Raw Material → Manufacturer.

Impact & Results

- A study estimates approximately 500–800 tonnes/day of plastic handled in the 13th Compound/13 Compounds area.
- This corresponds to an approximate annual scale of 187,000 tonnes based on the daily estimate.
- Main activities: collection, sorting, processing and manufacturing.

SWOT Analysis

Strengths: Strong material recovery, employment and entrepreneurial ecosystem.

Weaknesses: Informal nature and occupational/environmental concerns.

Opportunities: Formalization, cleaner technologies and circular manufacturing.

Threats: Regulatory pressure, unsafe processing and changing scrap markets.

Comparison with India

Dharavi shows the economic role of India's informal recycling sector. Formal systems can provide better environmental controls and worker protection.

Recommendations

- Recognize and integrate informal recyclers.
- Provide safer recycling technologies.
- Improve worker protection.
- Provide access to finance.
- Develop formal collection/recycling partnerships.

Conclusion

Dharavi demonstrates that informal recycling can recover significant material and generate livelihoods, while integration with formal environmental standards is important.

One-Page Exam Revision Sheet

Dharavi = Waste Pickers → Scrap Dealers → Sorting → Processing → Manufacturing → Circular Economy.

10. Dell's Circular Economy Strategy

Executive Summary

Dell has incorporated circular-economy principles into product design, materials and end-of-life management. Its strategy includes recycled and renewable materials, design for disassembly, responsible recycling and lifecycle services.

Background

Traditional electronics follow:

Raw Materials → Manufacturing → Consumer → Disposal.

Circular economy aims for:

Raw Materials → Manufacturing → Use → Recovery → Reuse/Recycling → New Products.

Strategy / Initiative

1. Sustainable materials.

2. Recycled plastics.

3. Closed-loop recycling.

4. Design for easier disassembly.

5. Repair/reuse/recycling.

6. Responsible packaging.

7. Lifecycle services.

Impact & Results

- Closed-loop plastics programme started in 2014.
- Recovered computer plastics can be converted into parts for new products.
- Dell reports 97.6% of packaging material as recycled/renewable.
- Product design considers easier disassembly and material recovery.

SWOT Analysis

Strengths: Circular product design, recycled materials and lifecycle approach.

Weaknesses: Complex supply chains and material tracking requirements.

Opportunities: Repair, refurbishment and material recovery.

Threats: Rapid technology changes and difficulty recovering products after consumer use.

Comparison with India

Dell uses product design, formal take-back/recycling and recycled materials. India faces challenges including informal collection and developing recycled-material markets.

Recommendations

- Design electronics for easier repair and disassembly.
- Increase take-back programmes.
- Use more recycled materials.
- Improve product traceability.
- Partner with authorized Indian recyclers.

Conclusion

Dell demonstrates how manufacturers can move from a linear make-use-dispose model toward a circular economy.

One-Page Exam Revision Sheet

Dell Circular Economy = Design → Use → Collect → Recover → Recycle → Reuse.

Super-Quick MSE-2 Master Revision Table

Case Study	Main Concept	Remember This
Japan	Organized waste management	Segregation + treatment
Switzerland	E-waste take-back	SENS + SWICO + ARF
India E-Waste Rules 2022	Regulation	EPR + Digital Portal
EPR	Producer responsibility	Target + Certificate + Recycler
Bengaluru	Urban e-waste problem	High generation + informal sector
Mumbai	Urban recycling ecosystem	Formal + informal network
Clean Kerala	Government waste model	Local government + HKS + CKCL
SWaCH Pune	Waste-picker integration	Cooperative + 4,000+ pickers
Dharavi	Informal recycling	Collection → processing → manufacturing
Dell	Circular economy	Reuse/recycle materials in products

Important Numbers to Remember

- Japan: 99.2% waste reduction rate; 19.3% recycling rate.
- Switzerland SENS: 100,762 tonnes processed in 2024; 11.14 kg/person.
- India E-Waste Rules: 106 EEE types, 7 categories.
- India EPR: 7,050 registered e-waste producers reported in 2024.
- Bengaluru: ~12,000 tonnes/year reported; ~15% scientifically processed in the cited report.
- Clean Kerala: 2,678 MT e-waste collected according to CKCL activity figures.
- SWaCH: 10 lakh households, 4,000+ waste-pickers, 82,891 tonnes recycled annually.
- Dharavi: approximately 500–800 tonnes/day plastic handled in the cited estimate.
- Dell: 97.6% of packaging reported as recycled/renewable material.

MSE-2 Answer Pattern

For a 5/10-mark case-study answer: Background → Strategy → 2–3 data points → SWOT → India comparison → Recommendations → Conclusion.

References

- Japan Ministry of the Environment – Municipal Solid Waste Statistics.
- SENS eRecycling – Facts and Figures 2024.
- Central Pollution Control Board – E-Waste Management Rules / EPR portal.
- Press Information Bureau – Government of India, EPR/e-waste information.
- Clean Kerala Company Limited – Official activity information.
- SWaCH Pune – Official website and impact information.
- Dharavi recycling research literature.
- Dell – Circular Economy and sustainability information.
- Mumbai Port Authority – E-waste management information.